Tools:SQL

PRESENTED BY SUSAN OFFIONG

WHAT WE KNOW ABOUT NEXTGEN CORP.

NextGen Corp. is a growing technology company that creates innovative software and hardware solutions.

The company focuses on hiring top talent and keeping employees satisfied to support its growth.

However, there are concerns about employee turnover, uneven performance, and salary differences among departments.

THE PLAN TO SUCCESS

Identify	Identify trends in employee retention and turnover to improve how we keep employees.
Track and assess	Track and assess performance in each department regularly.
Examine	Examine how salary relates to performance to ensure fairness and keep employees satisfied, while maintaining fair salary structures across all departments.

EMPLOYEE RETENTION ANALYSIS

Query Query History

```sql
--Q1.Who are the top 5 highest serving employees?

SELECT
    employee_id,
    first_name,
    last_name,
    hire_date,
    AGE(CURRENT_DATE, hire_date) AS tenure
FROM employee
ORDER BY hire_date
LIMIT 5;
```

Data Output Messages Notifications

Showing rows: 1 to 5 Pag

	employee_id [PK] integer	first_name character varying (30)	last_name character varying (30)	hire_date date	tenure interval
1	8	David	Moore	2015-06-30	10 years 8 mons 11 days
2	48	Frank	Smith	2015-09-08	10 years 6 mons 3 days
3	67	Jane	Brown	2015-10-12	10 years 4 mons 30 days
4	44	John	Johnson	2015-10-27	10 years 4 mons 15 days
5	57	John	Doe	2016-02-16	10 years 24 days

Total rows: 5 Query complete 00:00:00.339

Top 5 Highest Serving Employees

- This query shows the five employees with the longest tenure at NextGen Corp. It calculates how long each employee has been with the company using their hire date.
- We arrived at the answer by selecting employee details from the employee table, used AGE(CURRENT_DATE, hire_date) to compute tenure in years, months, and days.
- It shows that David Moore, Frank Smith, Jane Brown, John Johnson, and John Doe have more than 10+ years of service, which is a strong indicator of loyalty.

INSIGHTS & RECOMMENDATIONS

Implement retention incentives for long-serving employees (bonuses, leadership opportunities).

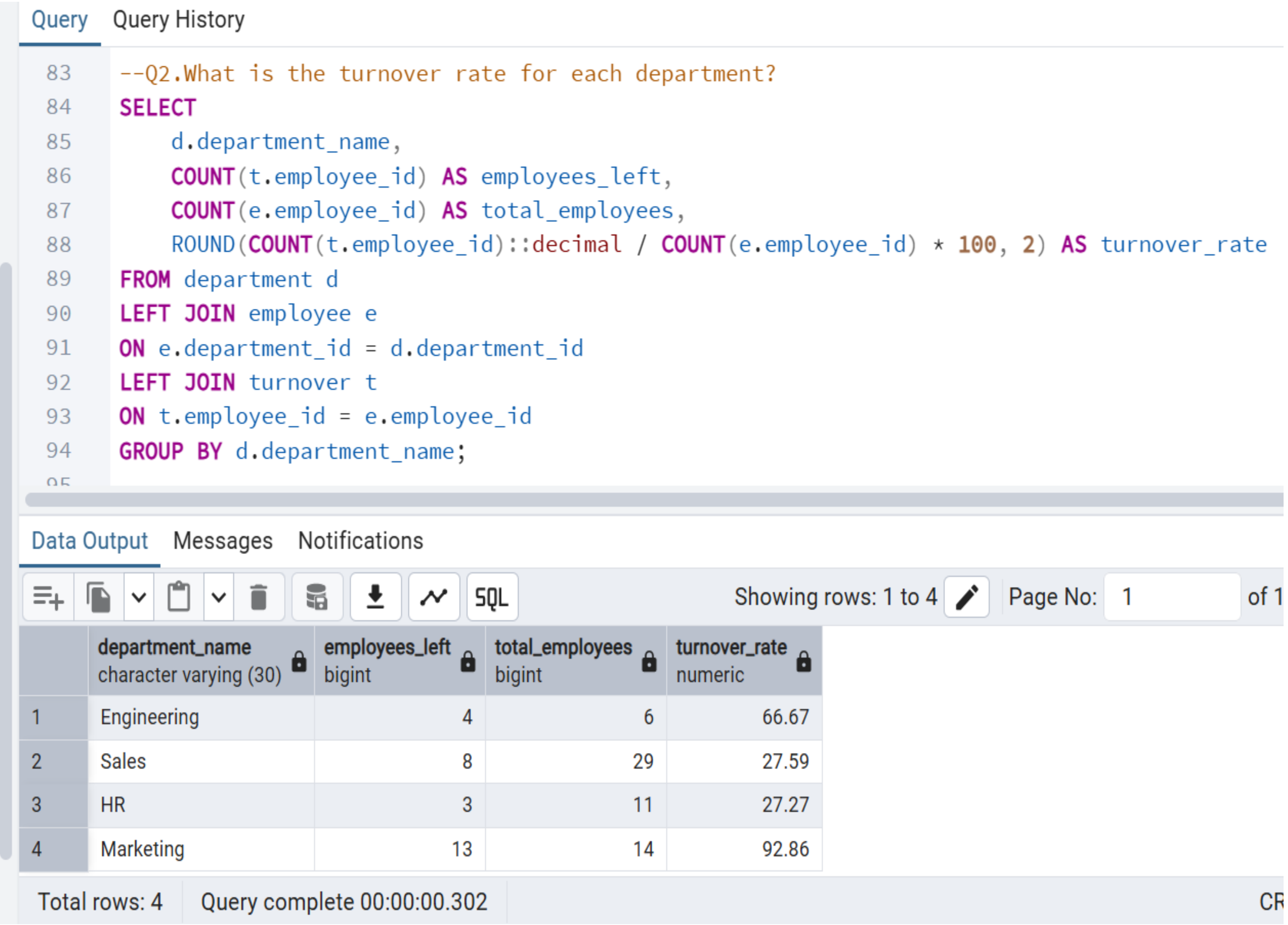
Conduct stay interviews to understand what keeps them loyal.

Use their experience to mentor newer employees and reduce turnover.

- Long-tenured employees are valuable assets; losing them would create knowledge gaps.
- Most long-serving employees were hired between mid-2015 and early 2016 — suggesting a strong hiring period.
- If turnover is rising now, it may indicate cultural or managerial changes compared to earlier years.

Turnover Rate for Each Department

- This output calculates the percentage of employees who left each department.
- It shows that Marketing has a turnover rate of 92.86%, Engineering: 66.67%, Sales & HR: ~27%.
- This output indicates severe retention issues in specific departments.


Query | Query History

```sql
--Q2.What is the turnover rate for each department?
SELECT
    d.department_name,
    COUNT(t.employee_id) AS employees_left,
    COUNT(e.employee_id) AS total_employees,
    ROUND(COUNT(t.employee_id)::decimal / COUNT(e.employee_id) * 100, 2) AS turnover_rate
FROM department d
LEFT JOIN employee e
ON e.department_id = d.department_id
LEFT JOIN turnover t
ON t.employee_id = e.employee_id
GROUP BY d.department_name;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

	department_name character varying (30)	employees_left bigint	total_employees bigint	turnover_rate numeric
1	Engineering	4	6	66.67
2	Sales	8	29	27.59
3	HR	3	11	27.27
4	Marketing	13	14	92.86

Total rows: 4 | Query complete 00:00:00.302

INSIGHTS & RECOMMENDATIONS

Marketing has a critical turnover problem — nearly the entire department left.

Engineering also shows high turnover (66%), which is dangerous for a tech company.

Sales and HR have moderate turnover, but still higher than ideal (industry average is 10–15%).

This suggests deeper issues in specific departments.

1

Conduct exit interviews focused on Marketing and Engineering to identify root causes.

2

Investigate the following: Leadership issues, Workload imbalance, Compensation dissatisfaction, Burnout, and Team conflicts.

3

Implement department-specific retention strategies like Better workload distribution, Manager training, Career development paths, and Competitive salary adjustments

Employees at Risk of Leaving (Based on Performance)

Showing rows: 1 to 60 | Page No: 1 of 1

	employee_id integer	first_name character varying (30)	last_name character varying (30)	department_name character varying (30)	avg_score numeric	latest_score numeric (2,1)	latest_score_date date
1	22	Grace	Wilson	Sales	3.43	3.0	2025-05-01
2	1	Jane	Wilson	Sales	3.58	4.1	2025-05-01
3	12	Eve	Davis	HR	3.65	3.0	2025-05-01
4	97	David	Smith	Engineering	3.72	3.2	2025-05-01
5	40	Alice	Wilson	HR	3.77	4.8	2025-05-01
6	16	Charlie	Lee	Sales	3.77	3.6	2025-05-01
7	42	Alice	Brown	Marketing	3.80	3.3	2025-05-01
8	28	John	Davis	Sales	3.83	4.5	2025-05-01
9	21	Grace	Smith	Sales	3.83	4.0	2025-05-01
10	36	Bob	Green	HR	3.85	4.4	2025-05-01
11	2	David	Lee	Sales	3.85	4.1	2025-05-01
12	85	David	Johnson	Sales	3.87	3.2	2025-05-01
13	66	Frank	Doe	Marketing	3.88	3.6	2025-05-01
14	8	David	Moore	Sales	3.90	3.0	2025-05-01
15	17	Frank	Green	Engineering	3.90	4.6	2025-05-01
16	98	Frank	Wilson	Marketing	3.92	4.7	2025-05-01

Total rows: 60 | Query complete 00:00:00.189 | CRLF | Ln 100

- This is a combined table that displays both the Average performance score (Dec 2024 → May 2025) and the Latest performance score (May 2025)
- This gives a complete view of both long-term consistency and recent performance trends.
- From the output, HR, Sales, and Marketing show mixed performance patterns.

Insights & Recommendations

Employees with avg_score < 3.5 may be long-term underperformers.

Employees whose latest_score < avg_score may be losing motivation and have a higher potential to leave the company.

Departments with many low averages may need training or leadership review.

Only six months of performance data is available for analysis, despite the company having employees who have worked for up to ten years.

Some departments have more low performers than others.

Investigate employees with declining recent scores; they may be disengaged.

Review department-level performance issues.

Reward high performers to improve morale and retention.

Provide targeted coaching for employees with low averages.

Main Reasons Employees Are Leaving

- This query outlines the most common reasons for employee turnover: Personal issues, finding another job, seeking career growth, and retirement.
- It also shows the total number of people who left for each specific reason.

Query Query History

```
--Q4.What are the main reasons employees are leaving the

SELECT
    reason_for_leaving,
    COUNT(*) AS total_count
FROM turnover
GROUP BY reason_for_leaving
ORDER BY total_count DESC;
```

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	reason_for_leaving text	total_count bigint
1	Personal	11
2	Found Another Job	7
3	Career Growth	5
4	Retired	5

Total rows: 4 Query complete 00:00:00.403

Insights & recommendations

- "Personal reasons" is the top reason reported by 11 individuals, yet the term remains vague and might conceal deeper issues.
- 7 employees left for better jobs, maybe competitors offer better pay.
- 5 employees left because they felt they had no career growth.
- 5 Retired from the total number of 28, which is normal but still contributes to turnover.

- Ask deeper questions during exit interviews to understand "personal reasons."
- Benchmark salaries against competitors.
- Create clearer career progression paths.
- Offer training and development to reduce career-growth exits.

PERFORMANCE ANALYSIS

```sql
--Performance Analysis
--Goal: Evaluate employee performance across different departments
--where performance can be improved.

--1.How many employees has left the company?
SELECT COUNT(DISTINCT employee_id) AS unique_employees_who_left
FROM turnover;
```

Data Output Messages Notifications

Showing rows: 1 to 1

	unique_employees_who_left bigint
1	28

How Many Employees Have Left The Company?

- **Out of 60 total employees, 28 employees left the company.**
- **This means almost half of the workforce (47%) exited.**
- **Most of these exits happened between 2024 and 2025, showing a clear spike in turnover during this period.**

Insights & Recommendations

A turnover rate of 47% is very high and signals a serious retention problem.

The fact that most exits occurred in 2024–2025 suggests something changed during this period, such As, Management issues, Salary dissatisfaction, Increased workload, Poor work environment, and better job opportunities elsewhere.

Losing this many employees in a short time can affect Productivity, Team stability, Company culture, and recruitment costs.

- INVESTIGATE WHY TURNOVER INCREASED SHARPLY IN 2024–2025.
- CONDUCT EXIT INTERVIEWS TO UNDERSTAND THE MAIN REASONS EMPLOYEES LEFT.
- IMPROVE SALARY, BENEFITS, AND CAREER GROWTH OPPORTUNITIES TO REDUCE FUTURE EXITS.
- STRENGTHEN EMPLOYEE ENGAGEMENT PROGRAMS TO IMPROVE SATISFACTION.
- TRAIN MANAGERS ON LEADERSHIP AND COMMUNICATION, AS POOR MANAGEMENT IS A COMMON CAUSE OF TURNOVER.
- MONITOR TURNOVER MONTHLY OR QUARTERLY TO DETECT EARLY WARNING SIGNS.

How many employees have a performance score of 5.0 / below 3.5?

- We combined the 6 months of scores (Dec 2024 → May 2025) to get **one average score per employee**.
- After calculating these averages, no employee reached the **highest performance level** (≥ 5.0), and only one employee fell into the **low-performance category** (< 3.5).
- This means most employees are performing in the **middle range**, not extremely high or extremely low.

Query | Query History

```
--2. How many employees have a performance score of 5
SELECT
    SUM(CASE WHEN avg_score >= 5.0 THEN 1 ELSE 0 END)
    SUM(CASE WHEN avg_score < 3.5 THEN 1 ELSE 0 END)
FROM (SELECT
        employee_id,
        AVG(performance_score) AS avg_score
    FROM performance
    GROUP BY employee_id) AS t;


SELECT employee_id, COUNT(*) AS months_recorded
FROM performance
GROUP BY employee_id;
```

Data Output | Messages | Notifications

SQL Showin

	employees_score_5_or_more bigint	employees_score_below_3_5 bigint
1	0	1

Insights & recommendations

- Performance was only measured for 6 months (from December 2024 to May 2025).
- Some employees, like David More, have been with the company since 2015, but there is no performance record for all those earlier years.
- This means the company did not track performance consistently before December 2024.
- Without earlier performance data, the company cannot see how employees performed over the years. Early signs of poor performance, Early signs of strong performance, Patterns that could help predict who might leave.
- The lack of long-term performance data makes it harder for HR to make good decisions about promotions, training, or retention.

- The company should start tracking performance continuously, not just for a short period.
- HR should create a standard performance review system that records scores every month or quarter.
- Older performance data should be collected or migrated into the system if possible.
- Managers should use performance trends to identify employees who need support early, reward high performers on time, and reduce turnover by addressing issues before they become serious.

```sql
--3. Which department has the most employees
SELECT
    d.department_name,
    SUM(CASE WHEN p.avg_score >= 5.0 THEN 1 E
    SUM(CASE WHEN p.avg_score < 3.5 THEN 1 EL
FROM (
    SELECT
        employee_id,
        AVG(performance_score) AS avg_score
    FROM performance
    GROUP BY employee_id
) p
```

a Output | Messages | Notifications

department_name character varying (30)	high_performers bigint	low_performers bigint
Sales	0	1
Engineering	0	0
HR	0	0
Marketing	0	0

Which department has the most employees with a performance of 5.0 / below 3.5?

- No department has any employee scoring 5.0 or above.
- Only one department (Sales) has one low-performing employee.
- All other departments have zero low performers.

Insights & Recommendations

- Sales may be facing High workload, Skill gaps, Low morale, and poor supervision
- The absence of high performers across all departments suggests a company-wide performance issue, not a department-specific one.

- Provide targeted training and support to the Sales department.
- Review workload, targets, and management style in Sales.
- Introduce performance improvement programs across all departments.
- Encourage managers to set clear goals and give regular feedback.

What is the Average Performance Score by Department?

- We calculated each employee’s average score across 6 months.
- We grouped these averages by department.
- We used ROUND(AVG(avg_score), 2) to show the department average to two decimal places
- All departments have similar performance levels, with Marketing slightly ahead.

```
--4. What is the average performance score by department?
SELECT
    d.department_name,
    ROUND(AVG(p.avg_score), 2) AS department_avg_performa
FROM (
    SELECT
        employee_id,
        AVG(performance_score) AS avg_score
    FROM performance
    GROUP BY employee_id
) p
JOIN employee e
```

Data Output Messages Notifications

Showing row

	department_name character varying (30)	department_avg_performance numeric
1	Marketing	4.17
2	HR	4.11
3	Sales	4.07
4	Engineering	4.01

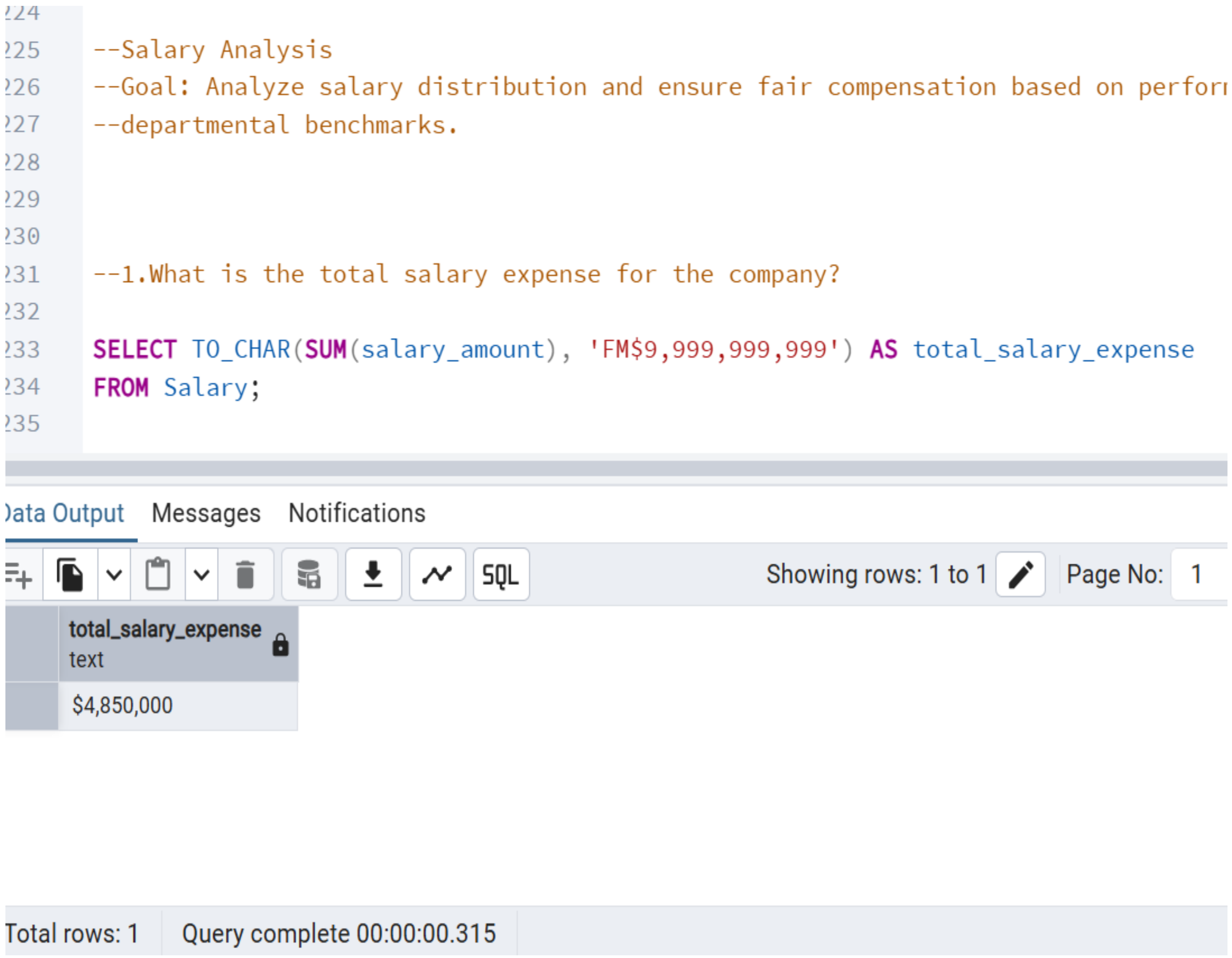
Total rows: 4 Query complete 00:00:00.201

Insights & Recommendations

- The performance gap between departments is small, meaning performance is fairly consistent across the company.
- Marketing and HR show slightly stronger performance.
- Engineering and Sales may need more support to improve their averages.
- The overall average scores (around 4.0) show moderate performance, not high performance.

- Support lower-scoring departments (Engineering and Sales) with training and coaching.
- Review manager effectiveness in departments with lower averages.
- Encourage knowledge-sharing between high-performing and lower-performing departments.
- Track performance continuously to identify long-term trends.

SALARY ANALYSIS



Total Salary Expense for the Company

- The output shows that the company spends $4.85 million annually on employee salaries.
- This represents the full payroll cost across all departments.

Insights & recommendations

- A payroll of $4.85 million indicates a mid-sized workforce with competitive salaries.
- In tech companies, payroll typically represents 60–70% of total operating expenses, significantly impacting profitability.
- High turnover can increase costs due to recruitment, training, and lost productivity.
- This figure also helps determine the feasibility of salary increases, potential bonuses, and staffing levels.

- Compare this salary expense with industry benchmarks to ensure competitiveness.
- If turnover is high, consider retention strategies — replacing employees is more expensive than retaining them.
- Conduct a year-over-year payroll trend analysis to detect salary inflation or over-hiring.
- Build a salary forecasting model to predict future payroll needs as the company scales.

Average Salary by Job Title

- This output presents the average salary for five different job titles within the company.
- This tells us how compensation differs across roles.

```
--2. What is the average salary by job title?

SELECT e.job_title,
       TO_CHAR(AVG(s.salary_amount), 'FM$99,999.00') AS avg_salary
FROM Employee e
JOIN Salary s
ON e.employee_id = s.employee_id
GROUP BY e.job_title;
```

Data Output Messages Notifications

SQL Showing rows: 1 to 5

	job_title character varying (30)	avg_salary text
1	HR Specialist	$81,818.18
2	Sales Representative	$84,285.71
3	Marketing Specialist	$77,857.14
4	Engineer	$80,000.00
5	Sales Manager	$80,000.00

Total rows: 5 Query complete 00:00:00.360

Insights & Recommendations

- Sales Representatives earn the highest average salary, likely due to revenue-driving responsibilities or commission structures.
- Marketing Specialists earn the lowest, which may indicate undervaluation or lower market rates.
- Engineers and Sales Managers have identical average salaries, suggesting a standardized pay band for mid-level roles.
- HR Specialists earn slightly above Engineers, showing HR is competitively compensated.

- Review whether Marketing Specialists are underpaid relative to workload and industry standards.
- Evaluate whether Sales Representatives' higher pay is tied to performance outcomes.
- Consider implementing salary bands to ensure fairness and transparency across roles.
- Conduct periodic salary audits to prevent pay inequities.

```sql
--3. How many employees earn above 80,000?

SELECT COUNT(*) AS employees_above_80k
FROM Salary
WHERE salary_amount > 80000;
```

Data Output | Messages | Notifications

	employees_above_80k bigint
1	26

Number of Employees Earning Above 80,000

- 26 out of 60 employees earn above $80,000.
- That means 43% of the entire workforce is in the high-earning bracket.
- This is a *very high proportion* for a company of this size.

Insights & Recommendations

- A high number of high-earning employees suggests a senior-heavy workforce and reliance on specialized roles, along with a competitive compensation strategy.
- Concentration of these employees in specific departments, like Sales or HR, may indicate uneven salary distribution and equity issues.
- This information helps HR identify critical talent and prioritize retention while assessing the budget impact of salary increases.

- Conduct a salary distribution analysis to see if high salaries align with high performance.
- Ensure that employees earning above $80k have clear KPIs tied to business outcomes.
- Ensure junior and mid-level staff have clear promotion pathways.

Performance–Salary Relationship Across Departments

This compares how well departments perform relative to how much they are paid.

```sql
--4. How does performance correlate with salary across departments?

SELECT d.department_name,
       TO_CHAR(AVG(s.salary_amount), 'FM$99,999.00') AS avg_salary,
       ROUND(AVG(p.performance_score), 2) AS avg_performance
FROM Employee e
JOIN Salary s ON e.employee_id = s.employee_id
JOIN Department d ON e.department_id = d.department_id
JOIN Performance p ON e.employee_id = p.employee_id
GROUP BY d.department_name;
```

Data Output Messages Notifications

SQL Showing rows: 1 to 4 Page N

	department_name character varying (30)	avg_salary text	avg_performance numeric
1	Engineering	$80,000.00	4.01
2	HR	$81,818.18	4.11
3	Sales	$82,068.97	4.07
4	Marketing	$77,857.14	4.17

Total rows: 4 Query complete 00:00:00.282

Insights & recommendations

- Marketing has the highest performance rating at 4.17 but the lowest salary, risking turnover in a small team.
- Sales has the highest salary without corresponding performance, indicating potential over-compensation.
- HR performs well at 4.11 with competitive pay, showing effective management and support.
- Engineering is stable but has slightly lower performance, possibly due to workload or skill gaps.

- Adjust Marketing compensation or introduce performance bonuses.
- Reevaluate Sales compensation structure — ensure pay is tied to measurable results.
- Provide Engineering with training or workload balancing.
- Implement a performance-based pay model to align salary with output across all departments.

RECOMMENDATION

1. Reduce Employee Turnover

- Focus on fixing issues in Marketing and Engineering where most employees are leaving.
- Talk to employees (exit + stay interviews) to understand what is causing dissatisfaction.
- Improve workload balance, manager support, and career growth to help people stay longer.

2. Improve Performance Management

- Set up a consistent performance review system (monthly or quarterly).
- Train managers to give clear goals and regular feedback.
- Support lower-scoring teams like Sales and Engineering with coaching and training.

3. Make Salary Fair and Linked to Performance

- Increase pay or rewards for high-performing teams like Marketing to prevent more resignations.
- Review Sales salaries to ensure pay matches actual performance.
- Create clear salary bands and performance-based bonuses to improve fairness and motivation

THANK YOU